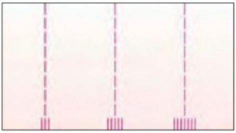
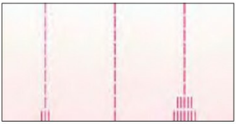
equally spaced. Another large array has been declared to hold the disks. The disks, shown as '|', should be arranged in odd manner, i.e., 1, 3, 5, 7, 9,… …17 (as 9 is the maximum height of our tower).

Function 'Start' is then declared. It checks whether the timer is 'on.' If the timer is not 'on,' the program initialises the array according to the current time and then creates a tower with random disks from the above chosen array of nine elements.

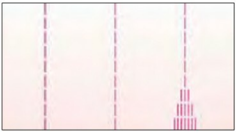
Another function called 'Hanoi' is declared. It takes the parameters 'from', 'to', 'intermediate' and 'tower height'. If the tower height is greater than 1, it calls the 'Hanoi' function and sends the current 'from,' 'to,' 'intermediate' and 'tower height' details. It then assigns the current 'from' item into the 'move from' array, increments the current disk position and places it in 'move to' array, and repeats the same until all the disks are completely and correctly placed onto the destination tower using the intermediate tower.


Screenshot 6: Moving the smaller disk from tower 'I' onto tower 'S'


Screenshot 7: Moving the large disk from tower 'I' onto tower 'D'

If height is not greater than 1, it places the source disk onto the destination disk without using any intermediary.

The 'tower' function creates the disks and places them on the 'S' tower. It also creates empty towers 'I' and 'D' and then calls the 'next tower' routine, which fires itself every second to scan for the current position according to the 'Hanoi' function.


Screenshot 8: All disks on tower 'D,' leaving 'S' and 'I' blank (solved)

For automatic operation, the program calls itself every five seconds after successful completion of the puzzle to reload another puzzle. The program calls the button via 'I' and 'D,' which in this case has been set to 'screen,' and projects, writes, paints or initialises the tower data onto it.

***EFY note.*** The source code of this article has been included in this month's EFY-CD and is also available on www.efymag.com.

## HANOI.HTM

```
<html>
<head>
<title>Automatic Hanoi Solver in JavaS-
cript</title>
<noscript style="font-family: Footlight
MT Light; font-size: 11pt; color: Sienna
;"> This Application Requires JavaScript
Plug-in which has to be initialized or
installed. It seems that your OS does not
have the Plug-in Installed or has been
disabled by you. Please Enable/Install
the Plug-in. If you have the plug-in and
is currently disabled, right-click on
the light yellow bar above, Chose "Allow
Blocked Content..." and then click "Yes"
from the consecutive dialog box, you should
see the Application running, or else in-
stall the plug-in from Internet/Reliable
Disc.</noscript>
<style>
.Hanoi
{
width:400px;
height:260px;
border:1;
font-family:Arial;
color:Deeppink;
font-size:15pt;
filter:progid:DXImageTransform.Micro-
soft.Gradient(startColorstr=Mintcream',
endColorstr='Mistyrose' GradientType=1);
text-align:center;
}
</style>
<script language="JavaScript">
var ht=9;
var t=null;
var mf=new Array(511);
var mt=new Array(511);
var tower=new Array(3);
var h=new Array(3);
var sp="          ";
var d=new Array ("|","|||","|||||","|||||
||","|||||||||","|||||||||||","||||||||||
|||","|||||||||||||||","|||||||||||||||||
","|||||||||||||||||||");
function Start()
{
if(!t)
{
now=new Date();
n=parseInt(now.getTime()/1000)%9+1;
delete now;
m=0;
Hanoi(0,2,1,n);
for(i=0;i<3;i++)
tower[i]=new Array(ht);
for(i=0;i<=n;i++)
tower[0][i]=n-i;
h[0]=n;
h[1]=h[2]=0;
im=0;
document.getElementById("Screen").
value=Tower();
t=window.setTimeout("Next_Tower()",1001);
}
}
function Hanoi(fr,to,bf,nm)
{
if(nm>1)
{
Hanoi(fr,bf,to,nm-1);
mf[m]=fr;
mt[m++]=to;
Hanoi(bf,to,fr,nm-1);
}
else
{
mf[m]=fr;
mt[m++]=to;
}
}
function Next_Tower()
{
tower[mt[im]][h[mt[im]]++]=tower[mf[im]]
[--h[mf[im]]];
document.getElementById("Screen").
value=Tower();
if(++im<m)
{
t=window.setTimeout("Next_Tower()",1001);
}
else
{
for(i=0;i<3;i++)
delete tower[i];
t=null;
}
}
function Tower()
{
s="\n";
for(i=ht-1;i>=0;i--)
{
for(j=0;j<tower.length;j++)
{
l=i<h[j]?tower[j][i]:0;
w=ht-l;
s+="  "+sp.substring(0,w)+d[l]+sp.
substring(0,w);
}
s+="\n";
}
return s;
}
window.setInterval("Start()",5001);
window.onload=Start;
</script>
</head>
<body oncontextmenu="return false"
onfocus="return false" onblur="return
false" onselectstart="return
false" ondragstart="return false"
ondragend="return false" class="Hanoi">
<form>
<input id="Screen" class="Hanoi"
type="Button" hideFocus onfocus="this.
blur()">
</form>
</body>
</html>
```